Finally, there is one consideration to make: in our experiment, we add completely random keys: since the range of values is so large, compared to the number of elements in the container, the expected number of duplicates is negligible, and we can assume that all sequence of keys to insert will be drawn with the same probability: in this scenario, it's unlikely that an adversarial sequence, causing the height to be super-logarithmic, will be chosen (the chances are beyond slim already for `n ~= 100`).

Basically, we are using the same concept of randomized treaps, moving the randomness into the generator of the sequences of keys to insert; unfortunately, however, we don't always get to choose the data to add to our containers!

And so, in order to verify this hypothesis, we can run a different experiment, reducing the influence of the random generator by limiting the set of possible keys; the results of limiting the keys to values in the range 0...1000 are shown in figure 3.18: now we can immediately notice a difference between the two data structures: the BST grows linearly, with a slope of approximately 10^{-3}, while the random treap still shows logarithmic height.

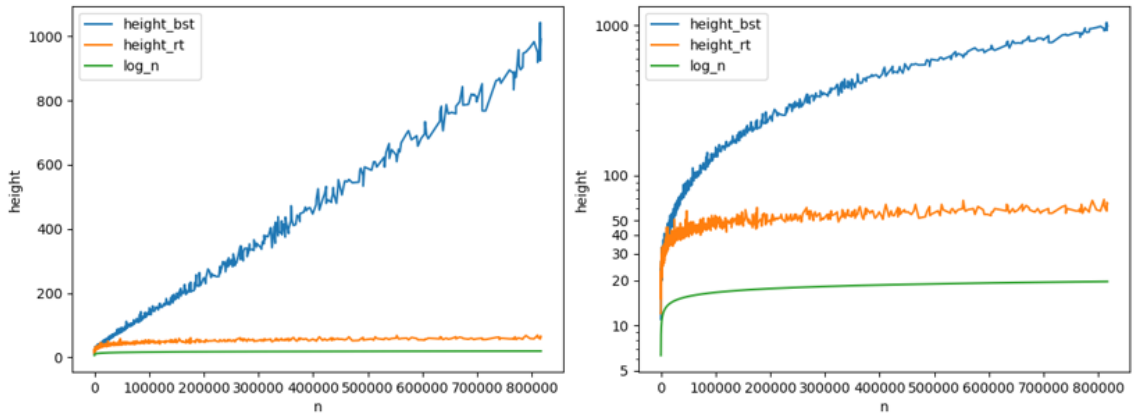


Figure 3.18 The height of a random treap versus a BST, when possible keys are randomly extracted from the set of integers between `0` and `1000`. The charts show the same